Jaesuk Kwak

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Objective and Interests

I am an B.S. student at the Department of Electrical and Computer Engineering at Seoul National University.
My research goal is to **bridge the gap between Embodied AI and Neuroscience**, particularly **improving data efficiency and cognitive inference capabilities in complex, long-horizon tasks.**
Specifically I aim to :

- **Boost Embodied AI with cognitive abilities** and intuitive understanding of the physical world, especially in the task of Spatial Navigation.
- **Build data-efficient AI** that mimics the human brain's learning mechanisms, enabling rapid adaptation from limited experiences.
- **Develop research tools to accelerate and ease labor-intensive neuroscience research**, so that valuable insights can be derived easily.

Keywords: Computer Vision, Reinforcement Learning, Computational Neuroscience, Cognitive Science.

Experience

- **Mandatory Military Service, Republic of Korea Army** — *September 2024 - March 2026*
  Signals Specialist — Defence Security Agency, Korea
- **SNU Computational Neuroscience Lab** — *January 2024 - September 2024*
  Research Intern, Advisor: Jeehyun Kwag — SNU College of Natural Science
  - Research on How Parvalbumin interneurons change network topology in the RSC.
  - Open-sourcing of Neural Topology Analysis a tool for visualizing topology from mouse brain in-vivo calcium imaging observations.
- **SNU Computer Vision Lab** — *July 2023 - January 2024*
  Research Intern, Advisor: Bohyung Han — SNU Samsung R&D Center
  - Developed Poisson Flow Generative Models via Image-charge based Diffusion.
  - Seminar on How Diffusion Models can be interpreted in a physical POV.
- **Sigma Intelligence Frontend Development** — *December 2022 - March 2023*
  Frontend Team Lead — Remote
  - Designed and launched Sigma Intelligence mobile application, reaching around 300 users under a week.

Education

- **Seoul National University** — *March 2022 - Expected February 2027*
  B.S. in Electrical and Computer Engineering — Seoul, Korea
  - GPA: 4.18/4.30
- **Seoul Science High School** — *March 2019 - February 2022*
  Secondary Education — Seoul, Korea
  - GPA: 4.23/4.30 (Top 10% in cohort)
  - Admitted one year early

Projects

- **Fine-tuning Navigation models with Policy Optimization unveils Internal Planners** — *March 2025 - Current*
  Tools: Deep Reinforcement Learning, Navigation Foundation Models, Diffusion Policy — []
  - Open-sourced headless data processing · evaluation · teleoperation pipeline for ROS based agents in Habitat-sim.
  - Novel contribution on RL training diffusion policy based navigation models.
- **Network Topology of Boundary Vector Cells in Mouse Retrosplenial Cortext** — *June 2024 - August 2024*
  Tools: Contrastive Learning, Spatial Navigation, Topological Data Analysis — []
  - Research on how Parvalbumin interneurons inhibit and change network topology in the Retrosplenial Cortex.
  - Awarded 1st place in 2024 Poster Presentation session by SNU College of Cognitive Sciences.
- **Neural Topology Analysis: Visualize network topology from images** — *June 2024 - August 2024*
  Tools: Contrastive Learning, Calcium Imaging — []
  - Open-sourced Neural Topology Analysis, an end-to-end pipeline for visualizing topology from calcium images.

- **Novel View Refinement via Attention Fusion** *March 2024 - June 2024*
  Tools: 3D Computer Vision, Stable Diffusion []
  - Open-sourced `Novel View Refinement`, generating unseen views of objects by fusing attention layers.
- **Fine-tuning code for SV3D** *March 2024 - June 2024*
  Tools: Fine-tuning, Stable Diffusion []
  - Open-sourced fine-tuning code for SV3D, the baseline model for `Novel View Refinement`.
- ***IMG-PFGM*: Diffusion Models Reconstructed as Image-charge Models** *June 2023 - Feburary 2024*
  Tools: Diffusion Models, Electromagnetics []
  - Proposed novel approach for interpreting diffusion models as minimizing total energy of electric fields between two charge distributions, and validated with toy model.
- **Non-invasive optogenetics with Ultrasonic Waves and Sonoactive Nanoparticles** *December 2023 - Feburary 2024*
  Tools: Optogenetics, Non-invasive surgery []
  - Proposed novel technology for non-invasive human optogenetics, awarded 1st place in a business competition for non-invasive surgery by SNU department of ECE.
- **Sigma Intelligence Mobile Application Development** *December 2022 - March 2023*
  Tools: NodeJS, Svelte []
  - Frontend team lead in developing an in-house mobile application for Sigma Intelligence affiliated with SNU ECE, reaching around 300 users in closed beta.
- **ROK Ministry of Defense Startup Competition projects** *March 2025 - September 2025*
  Tools: Presentation, Collaboration []
  - Awarded *Chief of Naval Operations Award* (Top 1% out of 440 teams).
  - As group lead, open-sourced projects and backend code for collaboration development.
    - *Mango*: AI Psychological Care for Military Personnel
    - *POLARIS*: RL-based mobile swarm drone real-time location identification system
    - *Milimeter*: A Map for soldiers with easier access to military benefits

Scholarships

- **Specialized Semiconductor Program Scholarship** *January 2024*
  SNU semiconductor program []
  - Selected as one of 200 students nationwide for a prestigious semiconductor talent program, awarded a scholarship of approximately 7,000 USD.
- **Presidential Science Scholarship** *May 2022*
  Korea Student Aid Foundation []
  - Full tuition and living expenses of approximately 30,000 USD. Awarded to top 60 students in the field every year. Field of Phyiscs.

Honors and Awards

- **Chief of Naval Officers Award** *June 2025*
  ROK Ministry of Defence Startup Competition []
  - Top 1% (out of 440 teams) with project *Milimeter*.
- **Certificate of Excellence** *September 2024*
  Chair of SNU College of Cognitive and Brain Sciences []
  - Awarded for excellence in research in SNU BCS 2024 Summer workshop.

Volunteer Experience

- **Lead of Organizing Team for 24w *GongDream* Engineering Contest** *January 2024 - February 2024*
  Society of Engineering Network and Service, SNU
  - Led the planning and execution of the engineering contest team as team lead for *Gong-Dream*, a STEM educational camp serving 300 high school students with 40 staff.
  - Funded by DB group and the SNU College of Engineering.
- **Lead of Organizing Team for 23s SNU Engineering Contest** *July 2023 - August 2023*
  Society of Engineering Network and Service, SNU
  - Led the planning and execution of the engineering contest team as team lead for SNU Summer Engineering Camp, a STEM educational camp serving 300 high school students with 40 staff.
  - Funded by the SNU College of Engineering.

- **STEM Educator in Engineering Outreach Program** *January 2023 - February 2023*
  Society of Engineering Network and Service, SNU
  - Served as an instructor in an engineering education initiative in the 2023 Winter *Gong-Dream*, a STEM educational winter camp for high school students.
  - Funded by DB group and the SNU College of Engineering.

Additional Information

Languages: English (full professional proficiency), Korean (Native proficiency)
Programming: Python, C++, Java, Svelte
ML frameworks: Pytorch, XGboost, HuggingFace
Design: Figma, Flutterflow